四年优化灌水日程：逐次日期、灌水量与五期汇总

2019 年

产量门槛=15.723 t·ha⁻¹；候选产量=15.725 t·ha⁻¹；季节实际灌水量=209.83 mm。

Table 2019. Optimized individual events.

Stage	Date	Actual depth (mm)
Seedling	2019-05-06	81.720754
Seedling	2019-05-07	3.447512
Jointing	2019-06-13	0.220281
Jointing	2019-06-14	4.224564
Jointing	2019-06-20	0.200626
Heading	2019-07-18	117.028543
Heading	2019-07-26	2.723275
Grain filling	2019-08-05	0.054565
Grain filling	2019-08-07	0.173138
Grain filling	2019-08-24	0.013362
Maturity	2019-09-01	0.024961
Maturity	2019-09-03	1.16e-09

Table 2019S. Five-stage totals.

Stage	Total depth (mm)	Positive events
Seedling	85.168267	2
Jointing	4.645471	3
Heading	119.751818	2
Grain filling	0.241065	3
Maturity	0.024961	2

2020 年

产量门槛=16.425 t·ha⁻¹；候选产量=16.427 t·ha⁻¹；季节实际灌水量=306.50 mm。

Table 2020. Optimized individual events.

Stage	Date	Actual depth (mm)
Seedling	2020-04-20	108.287640
Jointing	2020-06-10	0.174299
Jointing	2020-06-21	93.930726
Jointing	2020-07-01	0.599161
Heading	2020-07-16	2.160578
Heading	2020-07-25	0.092987
Heading	2020-07-26	0.001515
Grain filling	2020-07-28	87.916543
Grain filling	2020-07-29	0.790977
Grain filling	2020-08-17	0.003147
Maturity	2020-08-23	12.465925
Maturity	2020-09-02	0.034240
Maturity	2020-09-09	0.041215

Table 2020S. Five-stage totals.

Stage	Total depth (mm)	Positive events
Seedling	108.287640	1
Jointing	94.704186	3
Heading	2.255080	3
Grain filling	88.710668	3
Maturity	12.541380	3

2021 年

产量门槛=15.067 t·ha⁻¹；候选产量=15.068 t·ha⁻¹；季节实际灌水量=264.50 mm。

Table 2021. Optimized individual events.

Stage	Date	Actual depth (mm)
Seedling	2021-04-27	0.000663
Seedling	2021-05-23	0.002344
Jointing	2021-06-15	92.959220
Jointing	2021-06-28	85.822216
Jointing	2021-07-09	0.000563
Heading	2021-07-20	0.053827
Grain filling	2021-07-29	85.634129
Maturity	2021-08-21	0.008609
Maturity	2021-08-26	0.018403

Table 2021S. Five-stage totals.

Stage	Total depth (mm)	Positive events
Seedling	0.003008	2
Jointing	178.781999	3
Heading	0.053827	1
Grain filling	85.634129	1
Maturity	0.027012	2

2022 年

产量门槛=13.376 t·ha⁻¹；候选产量=13.390 t·ha⁻¹；季节实际灌水量=257.47 mm。

Table 2022. Optimized individual events.

Stage	Date	Actual depth (mm)
Seedling	2022-04-17	0.035146
Jointing	2022-06-02	0.109382
Jointing	2022-06-11	181.310872
Heading	2022-07-08	2.460366
Heading	2022-07-18	0.474977
Grain filling	2022-07-21	71.524972
Grain filling	2022-07-22	1.505109
Maturity	2022-08-13	0.044430

Table 2022S. Five-stage totals.

Stage	Total depth (mm)	Positive events
Seedling	0.035146	1
Jointing	181.420254	2
Heading	2.935343	2
Grain filling	73.030081	2
Maturity	0.044430	1